Hakancan Ozturk

hakancan.ozturk23@gmail.com • +44 7436010077 • linkedin/hakancan • github/hkc5 • hakancan.io

PROFESSIONAL EXPERIENCE

Software Development Engineer | ***Amazon*** *- London, UK* 01/2025 - Present

- Built Claude-powered agentic platform (won London-wide hackathon) analyzing 10M+ Prime Video metrics via MCP for real-time oncall investigation and CI/CD fix suggestions, triaging code changes, experiments, and device logs to auto-raise PRs, saving 100s of engineer-hours monthly across 1000+ engineers.
- Leading development of an LLM-based visual and functional UI defect detection system running 100s of times daily, catching bugs in the CI/CD pipeline before reaching production with 99% accuracy, replacing legacy rule-based tests.
- Migrated metric reporting subsystem from client-side to backend using Rust and AWS Lambda, processing 10K+ metrics per second while reducing app memory usage by 2%, delivering 7-figure annual cost savings.
- Drove AI tooling advocacy across the org, delivering multiple presentations and authoring internal wikis with 40K+ views to accelerate adoption of AI-powered developer workflows.

Founding Engineer | ***Albus Technologies*** *- London, UK* 03/2024 - 01/2025

- Engineered and optimized a scalable Retrieval-Augmented Generation (RAG) system with context-enriched vector search, boosting retrieval relevance by 30% and serving 100K+ customers.
- Developed scalable real-time Document Semantic Extraction pipelines and LLM agents using AWS (Lambda, S3, SQS, EC2), processing millions of PDF pages and over 50K minutes of audio.

Computing Researcher | ***Max Planck Institute*** *- Stuttgart, Germany* 06/2022 - 12/2022

- Enhanced robotics dynamics prediction by deploying ML data pipelines (SVMs, curve fits) analyzing 10TB simulation data; optimized CFD simulations via novel HPC scheduling, achieving 200x speedup, enabling 4 publications.

EDUCATION

Imperial College London | *MSc in Applied Computational Science and Engineering* 2023-2024

Highest overall grade in class | Class representative Distinction (78.27%)

Modules: Machine/Deep Learning, Numerical Methods, Computational Maths, Optimization

Koc University | *BSc in Mechanical Engineering* 2020-2023

Ranked 1st in class | Graduated one year early | Merit scholarship ($30k annually) GPA: 3.99/4.00

PROJECTS

MSc Dissertation: AI Surrogate Modeling for Turbulent Flow Simulations | *Imperial College London* 2024 - 2025

- Discovered a novel Grid-Invariant AI architecture combining convolutional autoencoders and adversarial networks to simulate high-fidelity turbulent flows, achieving unprecedented grid independence and scalability (PyTorch).
- Conducted 2000+ GPU hours of HPC for model optimization, enhancing long-term stability by 35% and prediction accuracy by 50%; open-sourcing advancements backed by NVIDIA (manuscripts in prep/review).

SKILLS & PUBLICATIONS

Languages & Libraries: Python, TypeScript, Rust, C++, MATLAB, PyTorch, scikit-learn, pandas, numpy, FastAPI, React

Cloud & DevOps: Docker, AWS, Google Cloud, Github Actions, Parallel Computing, High-Performance Computing

Publications | *5 papers with >30 citations*

- *Microrobotic Locomotion in Blood Vessels: A Computational Study on the Performance...* Advanced Intelligent Systems, 2023.
- *The Mismatch between Experimental and CFD Analyses for Magnetic Surface Microrollers* Nature Scientific Reports, 2023.
- *Learning-Enhanced 3D Fiber Orientation Mapping in Thick Cardiac Tissues* Optica Open, 2025.
- *Anisotropic Surface Microrollers for Endovascular Navigation: Computational Analysis* Advanced Theory and Simulations, 2025.
- *Locomotion Behavior of Magnetic Microrollers in Confined Tubular Geometries...* MARSS 2025, West Lafayette, USA.

ACHIEVEMENTS & AWARDS

- Y Combinator AI Startup School (San Francisco, 2025): Selected among top 2000 CS students/grads globally.
- Viridien & Imperial Hackathon Winner: Developed ML models for Carbon Capture & Seismic Data Analysis.
- High School Entrance Exam: Ranked 1st/1M; University Entrance Exam: Ranked 300th/2M (Turkey); IELTS: 8.0.